

ANIMAL MODELS VERIFY THE GENETIC BASIS FOR CUTANEOUS NEOPLASMS

Using human hereditary cancer syndromes as a guide, genetically engineered animal models have been developed that validate the contribution of individual genes to human cancer susceptibility through prospective experimental analysis. [42] These models have also contributed to the understanding of tumor initiation, progression, and the molecular mechanisms underlying cutaneous neoplasia.

Gene Targets for Mutations in Hereditary and Sporadic Cutaneous Cancers

GENE	FUNCTION	LOCUS	TUMOR TYPE	SYNDROME	SPONTANEOUS
TP53	DNA repair, apoptosis, cell cycle regulation	17p13.1	BCC, SCC	Li-Fraumeni (but no increase in skin cancers)	Yes
XPA, XPB		DNA repair	9q22.3, 2q21	BCC, SCC, melanoma	Xeroderma pigmentosum
XPC, XPD		DNA repair	3p25, 19q13.2-q13.33	BCC, SCC, melanoma	Xeroderma pigmentosum
XPF, XPG		DNA repair	16p13.3-p13.13, 13q33	BCC, SCC, melanoma	Xeroderma pigmentosum

Note: TP53 is commonly referred to as p53. The XPA–XPG genes are involved in nucleotide excision repair (NER), a critical pathway for removing UV-induced DNA damage. Defects in these genes underlie the extreme photosensitivity and high skin cancer risk in xeroderma pigmentosum (XP) patients.

The identification of mutations in key regulatory pathways—such as the **Hedgehog (Hh) signaling pathway** (*PTCH1*), **cell cycle control** (*INK4a-ARF* locus), and **DNA repair mechanisms**—has provided critical insights into the molecular pathogenesis of both hereditary and sporadic skin cancers. Animal models harboring targeted disruptions in these genes recapitulate human disease phenotypes, confirming their functional roles in tumorigenesis.

For example: - **PTCH1** knockout mice develop basal cell carcinomas, supporting its role as a tumor suppressor in the Hh pathway. - Mice with mutations in **p53** show increased susceptibility to UV-induced skin cancers. -

Models with deficiencies in **DNA repair genes** (e.g., *XPA*, *XPC*) exhibit heightened sensitivity to carcinogens and spontaneous tumor development, mirroring the XP phenotype.

These models not only confirm genetic causality but also serve as platforms for testing preventive strategies and targeted therapies.

Ongoing research into genetic susceptibility, gene-environment interactions, and somatic mutation profiles continues to refine our understanding of skin cancer etiology and opens avenues for precision medicine approaches in dermatologic oncology.